

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

5. Let $P = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2), (2, 3), (2, 4), (3, 1), (3, 2), (2, 3), (3, 4)\}$.

- (a) Illustrate the application of a divide-and-conquer algorithm to compute the closest pair of points in P .
- (b) Use the example to explain why the asymptotic cost is $O(n \log(n))$ for n points.

/15

Answer:

5. (a) We split the list of 12 points in two halves: $A = \{(1, 1), (1, 2), (1, 3), (1, 4), (2, 1), (2, 2)\}$, $B = \{(2, 3), (2, 4), (3, 1), (3, 2), (2, 3), (3, 4)\}$, and then compute $\delta_A = 1$ and $\delta_B = 1$ as the distance of the closest pairs respectively in A and B .

In the merge stage, we take $\delta = \min(\delta_A, \delta_B) = 1$, as the distance from the middle, which is the vertical line $x = 2$.

Starting at the point $(2, 1)$, we consider all points within distance 1 from $(2, 1)$: $(1, 1)$, $(2, 2)$, $(3, 1)$. And as all those three points are at distance 1, we find no new closest pair.

The minimal distance remains 1 as we continue to consider the points $(2, 2)$, $(2, 3)$, and $(2, 4)$. In each case, we need to examine respectively 4, 4, and 3 points, all at distance 1.

The first pair $\{(1, 1), (1, 2)\}$ at distance 1 is reported as the closest pair of points.

- (b) If $T(n)$ is the cost to compute the closest pair of n points, then we have the recurrence $T(n) = 2T(n/2) + O(n)$, because
- i. the size of the left and right halves A and B is $n/2$, and
 - ii. while all n points can be within the same distance from the middle line, we need to consider only a constant number of points as we examine each point close to the middle line, and the cost of the combination stage is therefore $O(n)$.